

Alexander Arias

Software Engineer

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SUMMARY

- Software Engineer with hands-on experience in hardware/software validation, automated testing, and CI/CD at Qualcomm.
- Strong foundation in systems programming, operating systems, and computer architecture using Python, C, C++, and Java.
- Builds systems end-to-end: a custom CPU/ISA from scratch and core OS subsystems (scheduling, concurrency, virtual memory).

WORK EXPERIENCE

Qualcomm — San Diego, CA

Engineering Technician – Operations Department (Software & Hardware Validation)

June 2025 – May 2026

- Executed Python-based automated test workflows (4–16 daily) across test benches supporting 10,000+ devices, modifying scripts under engineer direction to adjust test parameters and fix scripting issues, using Agile methodologies alongside Advantest and PMIC/RF systems.
- Troubleshoot hardware and software failures, analyzed diagnostic data, reproduced defects, and flagged invalid results requiring test stoppage and engineering review; documented findings in internal validation tools.
- Collaborated with 5–6 engineers through iterative defect reproduction, debugging, and fix-verification cycles to maintain hardware and software validation reliability.
- Advanced CI/CD validation cycles by executing automated tests across variable device batches, analyzing results, reproducing failures, and verifying fixes across evolving builds.
- Documented validation metrics, failure behavior, and troubleshooting findings across variable workloads while consistently exceeding monthly performance targets.

University of California San Diego — San Diego, CA

Computer Science Outreach Volunteer – Eta Kappa Nu Engineering Honor Society

January 2023 – December 2023

- Delivered STEM outreach to 100+ elementary and high school students weekly across three 7-week programs, introducing fundamental computer science concepts.
- Supported interactive coding and algorithm demonstrations for student engagement in computing topics.
- Mentored high school students on academic pathways in computer science and engineering.

Computer Science Tutor – CSE Department

September 2022 – December 2022

- Provided 200+ hours of one-on-one tutoring across a 10-week quarter, supporting 50+ students with programming and coursework-related challenges.
- Reinforced concepts in C, Assembly, and computer architecture through 5–15 minute tutoring sessions.

PROJECTS

Custom ISA and CPU — *SystemVerilog, Python, C, Git*

- Led a 3-person software engineering team in designing and implementing a custom 9-bit CPU architecture, including an 8-instruction ISA, 8 registers (R0–R7), instruction encoding, datapath, and program counter.
- Engineered and validated all 8 instructions through component and end-to-end testing, diagnosing hardware and machine-code behavior in Quartus and successfully executing required Hamming encoding, error correction, and pattern-recognition workloads 4 weeks ahead of schedule.

Nachos Operating System — *Java, C, Linux, Git*

- Implemented 3 operating-system projects as part of a 4-person software engineering team, covering multithreading and synchronization, process management and multiprogramming, CPU scheduling, and virtual memory.
- Designed a first-in, first-out (FIFO) CPU scheduling algorithm and contributed to locks, semaphores, condition variables, system calls, paging, page tables, page faults, and memory management; developed and executed tests to validate concurrent behavior, process execution, and virtual-memory functionality.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, C#, JavaScript, Assembly, SystemVerilog, HTML, CSS

Frameworks and Libraries: .NET, Node.js, JUnit, OpenGL

Software Engineering: Software Development Life Cycle (SDLC), Object-Oriented Programming & Design (OOP), Agile, CI/CD, Data Structures & Algorithms, Multithreading & Concurrency, Debugging, Software Testing, Unit Testing, Automated Testing

Development Tools: Git, GitHub, PyCharm, IntelliJ IDEA, Visual Studio, VS Code, GDB, Valgrind, Linux Command Line

Operating Systems: Linux, Windows

EDUCATION

University of California San Diego — San Diego, CA

Bachelor of Science in Computer Science

December 2023